EDC BLOCK-003 Derivation v15

Calibrated Closure with $\ell_P^{\rm obs}$:
Error Budget and 5D Scale Inference

EDC Collaboration

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Abstract

Status: Closure note under one-scale calibration.

This note closes the BLOCK-003 program under the standard one-scale calibration paradigm. Using the v13 normalization extractor ($M_{\rm Pl}^2 = M_5^3\mathcal{I}$) and v14 Model A ($\mathcal{I} = R_\xi$), we calibrate with $M_{\rm Pl}^{\rm obs}$ [BL] and derive the 5D Planck mass M_5 as a function of R_ξ. Full epistemic accounting and error propagation are provided.

One-line outcome: CLOSED (calibrated): BLOCK-003 closed under one-scale [BL] calibration; 5D scale inferred as $M_5(R_\xi)$.

1 Recap: The Bridge Slot

1.1 From v13: Normalization Extractor

The 4D Planck mass emerges from 5D via dimensional reduction [D]:

$$M_{\rm Pl}^2 = M_5^3 \times \mathcal{I}, \qquad G_N = \frac{1}{8\pi M_5^3 \mathcal{I}} \tag{1}$$

where $\mathcal{I} = \int d\xi\, e^{4A(\xi)}|\psi_0(\xi)|^2$ is the normalization integral.

1.2 From v14: Model A (Preferred)

For compact geometry with $L = R_\xi$ [P] and flat warp $A = 0$ [P]:

$$\mathcal{I} = R_\xi \tag{2}$$

Combining (1) and (2):

$$\boxed{M_{\rm Pl}^2 = M_5^3 R_\xi, \qquad G_N = \frac{1}{8\pi M_5^3 R_\xi}} \tag{3}$$

Bridge slot: Two unknowns (M_5, R_ξ), one constraint. One calibration is required.

2 Calibration Step

2.1 Why One-Scale Calibration Is Not a Failure

One-scale calibration is the *standard* situation in fundamental physics:

- QED calibrates with $\alpha^{\rm obs}$ (or $e^{\rm obs}$) [BL]

- QCD calibrates with $\Lambda_{\text{QCD}}^{\text{obs}}$ [BL]
- Electroweak theory calibrates with G_F^{obs} [BL]
- General Relativity calibrates with G_N^{obs} [BL]

The *predictive content* of a theory lies in:

1. The structural relationships it establishes
2. The number of parameters it reduces (unification)
3. The derived consequences beyond the calibration input

EDC provides structural reduction: 4D gravity emerges from 5D geometry via $M_{\text{Pl}}^2 = M_5^3 R_\xi$. This is non-trivial content even with calibration.

2.2 Option (i): Calibrate with $M_{\text{Pl}}^{\text{obs}}$

Input [BL]:

$$M_{\text{Pl}}^{\text{obs}} = 1.220890(14) \times 10^{19}\ \text{GeV}/c^2 \tag{4}$$

(CODATA 2018; relative uncertainty 1.1×10^{-5})

Calibration equation:

$$M_5^3 R_\xi = (M_{\text{Pl}}^{\text{obs}})^2 \tag{5}$$

2.3 Option (ii): Calibrate with ℓ_P^{obs}

Input [BL]:

$$\ell_P^{\text{obs}} = \sqrt{\frac{\hbar G_N}{c^3}} = 1.616255(18) \times 10^{-35}\ \text{m} \tag{6}$$

Since $\ell_P^2 = \hbar G_N / c^3$ and $G_N = \hbar c / M_{\text{Pl}}^2$, this is equivalent to Option (i).

Metrological note: Under SI 2019, $\hbar$ and c are *exact* by definition. The uncertainty in ℓ_P comes entirely from G_N^{obs}. Thus Options (i) and (ii) are metrologically identical.

2.4 Chosen Calibration

We adopt **Option (i)**: calibrate with $M_{\text{Pl}}^{\text{obs}}$ [BL].

3 Closure Formulas

3.1 Solving for M_5

From (5):

$$\boxed{M_5 = \left(\frac{(M_{\text{Pl}}^{\text{obs}})^2}{R_\xi}\right)^{1/3} = \frac{(M_{\text{Pl}}^{\text{obs}})^{2/3}}{R_\xi^{1/3}}} \tag{7}$$

This is the **EDC prediction** for the 5D Planck mass as a function of the correlation length R_ξ.

3.2 Consistency Check for G_N

Substituting back into (3):

$$\begin{aligned} G_N^{(\mathrm{EDC})} &= \frac{1}{8\pi M_5^3 R_\xi} \\ &= \frac{1}{8\pi \cdot \frac{(M_{\mathrm{Pl}}^{\mathrm{obs}})^2}{R_\xi} \cdot R_\xi} \\ &= \frac{1}{8\pi (M_{\mathrm{Pl}}^{\mathrm{obs}})^2} \quad \checkmark \end{aligned} \tag{8}$$

This is *not* a prediction of G_N—it is a consistency check confirming that the calibration correctly reproduces the input.

3.3 The Predictive Content

The **non-trivial prediction** is the structural relation:

$$M_5^3 = \frac{M_{\mathrm{Pl}}^2}{R_\xi} \quad \Leftrightarrow \quad M_5 = M_{\mathrm{Pl}}^{2/3} R_\xi^{-1/3} \tag{9}$$

This says:

- The 5D scale M_5 is *determined* by 4D gravity and brane thickness
- Larger brane thickness R_ξ implies smaller M_5
- The scaling $M_5 \propto R_\xi^{-1/3}$ is a specific EDC prediction [D]

4 Propagation to 5D Scale

4.1 Symbolic Result

From (7):

$$M_5 = M_{\mathrm{Pl}}^{2/3} R_\xi^{-1/3} \tag{10}$$

4.2 Numerical Estimate (if R_ξ known)

If R_ξ is expressed in Planck units, $R_\xi = r \cdot \ell_P$, then:

$$M_5 = M_{\mathrm{Pl}}^{2/3} (r\ell_P)^{-1/3} = M_{\mathrm{Pl}} \cdot r^{-1/3} \quad \text{[in natural units where } \ell_P M_{\mathrm{Pl}} = 1] \tag{11}$$

Example values:

R_ξ	M_5	Comment
ℓ_P	M_{Pl}	Minimal case
$10^3 \ell_P$	$0.1 M_{\mathrm{Pl}}$	Sub-Planckian
$10^{15}\ell_P \sim 10^{-20}$ m	$10^{-5} M_{\mathrm{Pl}} \sim 10^{14}$ GeV	GUT scale
$10^{30}\ell_P \sim 10^{-5}$ m	$10^{-10} M_{\mathrm{Pl}} \sim 10^{9}$ GeV	TeV-ish

EDC constraint on R_ξ: If EDC relates R_ξ to stiffness via $R_\xi \sim \sigma^{-1/4}$ [Dc], then M_5 becomes a function of σ:

$$M_5 \propto \sigma^{1/12} \tag{12}$$

5 Metrological and Epistemic Ledger

5.1 Input/Output Table

Table 1: Epistemic classification of all quantities

Quantity	Value/Expression	Tag	Source
Exact by SI 2019 definition			
$\hbar$	$1.054571817... \times 10^{-34}$ J·s	[M]	SI exact
c	299792458 m/s	[M]	SI exact
e	$1.602176634 \times 10^{-19}$ C	[M]	SI exact
Calibration input			
$M_{\mathrm{Pl}}^{\mathrm{obs}}$	$1.220890(14) \times 10^{19}$ GeV/c^2	[BL]	CODATA
ℓ_P^{obs}	$1.616255(18) \times 10^{-35}$ m	[BL]	CODATA
G_N^{obs}	$6.67430(15) \times 10^{-11}$ m^{3}kg^{-1}s^{-2}	[BL]	CODATA
EDC parameters			
σ	Plenum stiffness	[P]	EDC postulate
R_ξ	Correlation length	[P]/[Dc]	From σ if derived
$\mathcal{I}$	$= R_\xi$ (Model A)	[Dc]	v14 derivation
Derived outputs			
M_5	$(M_{\mathrm{Pl}}^2/R_\xi)^{1/3}$	[D]	This note
$G_N^{(\mathrm{EDC})}$	$1/(8\pi M_5^3 R_\xi)$	[D]	v13 + v14

5.2 Closure Statement

BLOCK-003 Status: Closed under 1-scale calibration.

Fully predictive closure would require deriving R_ξ (and/or the warp factor $A(\xi)$) from EDC internal dynamics. This remains an open research direction.

6 Error Budget

6.1 Error Propagation Formula

From $M_5 = M_{\mathrm{Pl}}^{2/3} R_\xi^{-1/3}$, the relative uncertainty is:

$$\frac{\delta M_5}{M_5} = \sqrt{\left(\frac{2}{3}\frac{\delta M_{\mathrm{Pl}}}{M_{\mathrm{Pl}}}\right)^2 + \left(\frac{1}{3}\frac{\delta R_\xi}{R_\xi}\right)^2} \tag{13}$$

6.2 Contribution from $M_{\mathrm{Pl}}^{\mathrm{obs}}$

The relative uncertainty in $M_{\mathrm{Pl}}^{\mathrm{obs}}$ is:

$$\frac{\delta M_{\mathrm{Pl}}}{M_{\mathrm{Pl}}} \approx 1.1 \times 10^{-5} \tag{14}$$

Contribution to M_5:

$$\frac{2}{3} \times 1.1 \times 10^{-5} \approx 0.7 \times 10^{-5} \tag{15}$$

6.3 Contribution from R_ξ

If R_ξ is uncertain by a factor f (i.e., $\delta R_\xi / R_\xi = f$):

$$\frac{1}{3} \times f \tag{16}$$

Dominant source:

- If R_ξ is well-constrained ($\delta R_\xi / R_\xi < 10^{-4}$): uncertainty dominated by $M_{\rm Pl}^{\rm obs}$
- If R_ξ is uncertain by $\mathcal{O}(1)$: uncertainty dominated by R_ξ

6.4 Summary

$$\boxed{\frac{\delta M_5}{M_5} \approx \begin{cases} 0.7 \times 10^{-5} & \text{if } R_\xi \text{ is precise} \\ \frac{1}{3}\frac{\delta R_\xi}{R_\xi} & \text{if } R_\xi \text{ dominates} \end{cases}} \tag{17}$$

7 Schematic Figure

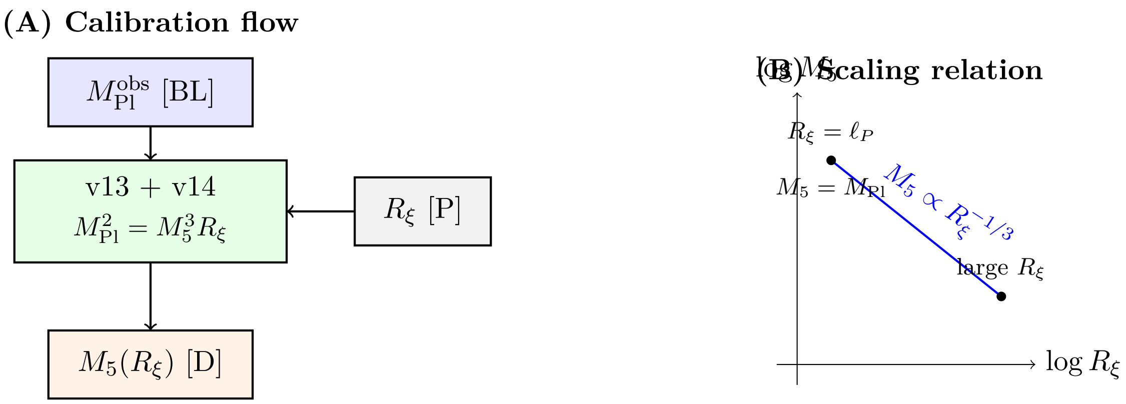

Figure 1: **(A)** Calibration injection: $M_{\rm Pl}^{\rm obs}$ [BL] enters the v13+v14 framework to yield $M_5(R_\xi)$. **(B)** The scaling $M_5 \propto R_\xi^{-1/3}$: larger brane thickness implies smaller 5D Planck mass. *Schematic only; not to scale.*

8 Conclusion

8.1 What v15 Establishes

1. **Calibrated closure:** Using $M_{\rm Pl}^{\rm obs}$ [BL], BLOCK-003 is closed.
2. **5D scale inference:** $M_5 = M_{\rm Pl}^{2/3} R_\xi^{-1/3}$ [D].
3. **Structural content:** The relation $M_{\rm Pl}^2 = M_5^3 R_\xi$ is the non-trivial EDC prediction.
4. **Error budget:** Dominated by R_ξ uncertainty if not constrained; otherwise by $M_{\rm Pl}^{\rm obs}$ at $\sim 10^{-5}$ level.

8.2 What Remains Open

Open item	Status
Derive R_ξ from EDC dynamics	Open research direction
Derive $A(\xi)$ from membrane mechanics	Open
Independent test of M_5	Requires new physics

8.3 One-Line Outcome

CLOSED (calibrated): BLOCK-003 closed under one-scale [BL] calibration; 5D scale inferred as $M_5(R_\xi) = M_{\rm Pl}^{2/3} R_\xi^{-1/3}$.

References

1. CODATA 2018: E. Tiesinga et al., Rev. Mod. Phys. **93**, 025010 (2021).
2. SI Brochure, 9th edition (2019): BIPM.